



Neobehaviorism, Radical Behaviorism, and Problems of Behaviorism

WHILE MANY PSYCHOLOGISTS ACCEPTED WATSON'S RHETORIC OF prediction and control, few accepted the theoretical details of his behaviorist system, and in the 1930s and 1940s the neobehaviorism of Clark L. Hull (1884–1952) and Edward C. Tolman (1866–1959) superseded Watson's positivist brand of behaviorism. Hull and Tolman followed Watson in maintaining that scientific psychology should be directed to the explanation, prediction, and control of observable behavior rather than introspected mental states, and they rejected the form of structural psychology championed by Titchener and his followers. Yet in contrast to Watson, they recognized the legitimacy of theoretical explanations of observable behavior in terms of the internal states of organisms, including their mental states, on a par with theoretical explanations of the observable properties of physical elements in terms of their internal composition and structure (such as the explanation of the properties of carbon in terms of its molecular composition and structure).

The neobehaviorist attempt to more closely approximate the theoretical orientation of the natural sciences marked an advance over Watson's restriction of behaviorist psychology to the description of observable stimulus-response sequences. However, in their attempt to model behaviorist psychology upon the natural sciences, neobehaviorists did not look to the actual practice of natural sciences such as physics and chemistry, but adopted the equally restrictive logical positivist account of theory advanced by philosophers of science in the early decades of the 20th century. As Sigmund Koch put it,

In pursuit of these ends, psychology did not go directly to physics, but turned instead for its directives to middlemen. These were, for the most part, philosophers of science (especially logical positivists) and a number of physical scientists who had been codifying a synoptic view of the nature of science and who, by the early

thirties, were actively exporting that view from their specialties in the scholarly community at large.

—(Koch, 1964, p. 10)

The logical positivist account of theory exercised a debilitating influence on the development of neobehaviorist theory. Burrhus F. Skinner (1904–1990) exploited the limitations of this account of theory and developed a **radical behaviorism** that eschewed theories about the internal states of organisms. Skinner returned to Watson's positivist conception of behaviorist psychology as restricted to the description, prediction, and control of observable behavior.

Neobehaviorism was the dominant orientation in American departments of psychology until after the end of the Second World War, when it was displaced by radical behaviorism. In the decades after the war, the theories of conditioned learning that provided the theoretical foundation for all forms of behaviorism came under increasing criticism, preparing the way for the cognitive revolution of the 1950s and 1960s. The decades following the Second World War also witnessed a major transformation of the institutional structure of scientific and professional psychology.

NEOBHAVIORISM

Watson's brand of behaviorism was an expression of his commitment to a positivist science restricted to the description, prediction, and control of observable behavior:

The behaviorist asks: Why don't we make what we can *observe* the real field of psychology? Let us limit ourselves to things that can be observed, and formulate laws concerning only those things. Now what can we observe? Well, we can observe behavior—what the organism does or says.

—(1924, p. 6)

It was also an expression of his *inductivist* conception of the development of scientific theories, according to which the aim of science is to accumulate observational laws that can eventually be integrated by summative theories, in the fashion in which the various gas laws relating pressure, temperature, and volume (Boyle's law, Charles's law and Graham's law) were independently established and then integrated by the kinetic theory of gases. As Watson put it,

You will find, then, the behaviorist working like any other scientist. . . . We collect our facts from observation. Now and then we select a group of facts and draw certain general conclusions about them. In a few years as new experimental data are gathered by better methods, even these tentative general conclusions have to

be modified. . . . Experimental technique, the accumulation of facts by that technique, occasional tentative consolidation of these facts into a theory or a hypothesis describe our procedure in science. Judged upon this basis, behaviorism is a true natural science.

—(1924, pp. 18–19)

In contrast, neobehaviorists such as Hull and Tolman embraced the *hypothetico-deductive* account of the development of scientific theories. According to this account, theories play more than a merely summative role. Theoretical postulates are introduced not only to accommodate previously established observational laws, but also to generate new observational predictions. Bohr's theory of the atom, for example, not only accommodated the known empirical laws governing spectral emissions, but also generated additional predictions about the spectral emissions of more complex elements and phenomena such as the Zeeman effect (the splitting of spectral lines in a magnetic field).

Thus Hull, for example, maintained that scientific psychology should proceed via the deduction of the empirical implications of postulated theoretical principles:

The typical procedure in science is to adopt a postulate tentatively, deduce one or more of its logical implications concerning observable phenomena, and then check the validity of the deductions by observation.

—(1943a, p. 15)

Similarly, if somewhat more picturesquely, Tolman famously proclaimed,

I in my future work, intend to go ahead imagining how, if I were a rat, I would behave.

—(1938, p. 24)

This was not intended as a return to speculative anthropomorphism, but as a heuristic for generating hypotheses about the cognitive states of animals, which could then be tested via their empirical implications.

The logical positivists developed the hypothetico-deductive account of scientific method into a general account of theoretical explanation in science. They conceived of explanation as the deduction of empirical laws (or theorems) from theoretical postulates (or axioms), modeled upon the deductive systems of Euclid and Newton. This account “held forth an ideal of rigorous theory and seemed to define a route towards its achievement”:

In barest outline, it asserts theory to be a hypothetic-deductive system. Laws or hypotheses believed fundamental are asserted as postulates, and the consequences of these (theorems) are deduced by strict logical and mathematical rules. The theorems

are then to be tested by experiment. Positive results increase the probability of a hypothesis; negative results call them into question.

—(Koch, 1962, p. 401)

Thus the difference between the behaviorism of Watson and the neobehaviorism of Hull and Tolman was essentially the difference between original positivism (or dogmatic empiricism) and logical positivism.

Logical Positivism

A group of philosophers, scientists, mathematicians, and social scientists who met in Vienna in the 1920s and early 1930s formulated the set of doctrines known as **logical positivism**, which played an influential role in the development of neobehaviorism. Moritz Schlick (1882–1936) founded the group that came to be known as the Vienna Circle, whose members included Gustav Bergmann (1906–1987), Rudolf Carnap (1891–1970), Herbert Feigl (1902–1988), Otto Neurath (1882–1945), and Kurt Gödel (1906–1978). The logical positivists were influenced by earlier forms of empiricism and positivism and by more recent developments in logic and science, notably Russell and Whitehead's reduction of arithmetic to logic in *Principia Mathematica* (1910) and Einstein's theory of relativity.

The central doctrines of logical positivism were essentially linguistic restatements of the psychological and meaning empiricism of David Hume (whom the positivists admired). Hume had claimed that the only meaningful ideas and terms are those derived from experience, and the logical positivists advanced the **verification principle**, according to which the only meaningful factual propositions are those verifiable by observation. As Moritz Schlick put it, "the meaning of a proposition is its method of verification" (Schlick, 1936, p. 341). The logical positivists distinguished between formal and factual propositions, between propositions rendered true or false by internal relations of meaning or logic (such as "all triangles have three sides") and propositions rendered true or false by facts about the world (such as "all unsupported bodies fall to the ground"), which corresponded to Hume's distinction between "relations of ideas" and "matters of fact and existence." They also appropriated Hume's account of causality in terms of the constant conjunction (or correlation) of observables.

As Hume had employed the principle of meaning empiricism to dismiss metaphysical ideas or concepts as "mere sophistry and illusion," the logical positivists employed the verification principle to dismiss metaphysical, ethical, religious, and aesthetic propositions as literal nonsense, because they could not be empirically verified. However, they recognized that many of the theoretical propositions of science, such as propositions about electrons and gravitational fields, were also incapable of direct empirical verification, but they were reluctant to dismiss them as nonsense. To accommodate theoretical propositions about unobservable entities

such as electrons and gravitational fields, the logical positivists claimed that it is legitimate to introduce theoretical propositions so long as they are defined in terms of observables, via **correspondence rules** or **operational definitions** relating theoretical propositions to propositions about observables.

In the early version of logical positivism known as **sensationalism**, observational propositions were held to describe the properties of private sense experience, such as the intensity of colors or apparent differences in weight. In the later version of logical positivism known as **physicalism**, observational propositions were held to describe the publicly observable properties of physical objects, such as readings on spectrometers or the motion of bodies. Neobehaviorists such as Hull and Tolman embraced this later version of logical positivism, which became known as **scientific empiricism** (or logical empiricism). They treated theoretical references to consciousness and cognition in psychology as analogous to theoretical references to electrons and gravitational fields in natural science, as theoretical postulates related to publicly observable behavior:

They are to behavior as electrons, waves, or whatever it may be are to the happenings in inorganic matter.

—(Tolman, 1932, p. 414)

S. S. Stevens, in “The Operational Definition of Psychological Concepts,” maintained that a theoretical proposition “has meaning . . . if, and only if, the criteria of its applicability or truth consist of concrete operations that can be performed” (1935, pp. 517–518). Scientific empiricists treated theoretical postulates relating to the internal states of organisms as **intervening variables**, defined in terms of relations between observable independent variables such as environmental or physiological stimuli and observable dependent variables such as behavioral responses (Bergmann & Spence, 1941; Carnap, 1936, 1937; Pratt, 1939). This conception of psychological theories held powerful sway during the heyday of neobehaviorism, from the 1930s to the 1950s, and remained influential for many subsequent decades.

The concept of an intervening variable was ambiguous, since it could be interpreted as either a **logically** or a **causally** intervening variable. According to the former interpretation, an intervening variable is merely a logical device for integrating descriptions of observable stimuli and behavioral responses. By the latter interpretation, an intervening variable represents an internal state of an organism that causally mediates between observable stimuli and behavioral responses. (Of course every causally intervening variable is also a logically intervening variable, but not every logically intervening variable need be treated as a causally intervening variable.) The logical interpretation of intervening variables was an expression of the instrumentalist conception of scientific theories, according to which theoretical postulates are merely logical devices for integrating observational laws,

whereas the causal interpretation was an expression of the realist conception of scientific theories, according to which theoretical postulates are potentially true descriptions of entities such as electrons and cognitive states.

Many neobehaviorists embraced an instrumentalist conception of theory and maintained that intervening variables merely serve as “economical devices to order experimental variables in relation to the dependent variables”:

They are “shorthand” descriptions, and nothing more, of the influence on behavior of several independent variables. The only meaning possessed by these intervening variables is their relationship to both the independent and dependent variables.

—(Kendler, 1952, p. 271)

Other neobehaviorists, notably Hull and Tolman, were committed realists who treated intervening variables as explanatory references to internal states that causally mediate between observed stimuli and behavior. However, they all insisted that intervening variables must be operationally defined in terms of observable stimuli and behavioral responses (even if they did not always practice their operational preaching).

The degree to which logical positivism influenced the development of Hull’s and Tolman’s original theoretical positions is a matter of dispute. Laurence D. Smith (1986) has argued that they developed their theoretical views independently of logical positivism and only later appealed to the scientific empiricist account as a convenient justification of their positions. However, their later metatheoretical and methodological pronouncements were clearly shaped by scientific empiricism (as Smith acknowledged), as was the theory and practice of later neobehaviorists such as Kenneth W. Spence (1907–1967) and Charles E. Osgood (1916–1991).

Operationism

The scientific empiricist account of the operational definition of scientific theories was supposedly based upon the **operationism** of Percy Bridgman (1882–1961), the Harvard physicist who won the Nobel Prize in physics in 1946 for his work on matter under high pressure. According to Bridgman (1927), any legitimate scientific concept must be linked to measurement procedures that can be employed to determine its empirical values. Such **operational measures** provide the empirical significance of a scientific concept or an operational definition of the concept. Bridgman claimed that since concepts such as “absolute simultaneity” and “absolute space” could not be operationally defined, they lacked empirical significance and were of no use in science.

However, Bridgman’s concern was with the scientific utility of concepts, not their theoretical meaning—thus a concept such as absolute simultaneity might be

perfectly meaningful but useless in science. Bridgman was critical of neobehaviorists who appropriated his operationism in support of their claim that theoretical propositions in psychology must be operationally defined in terms of observable stimuli and behavior (Koch, 1992), and in later years lamented that he had “created a Frankenstein” (cited in Green, 1992, p. 310).

Edward C. Tolman: Purposive Behaviorism

Edward Chance Tolman was born in Newton, Massachusetts, in 1886. Edward and his brother Richard earned undergraduate degrees in electrochemistry at MIT, where their father had been a member of the first graduating class. Richard went on to a distinguished career in chemistry. Edward became interested in philosophy and psychology after reading James's *Principles* during his last year at MIT.

Tolman enrolled at Harvard in 1911 with the aim of studying both philosophy and psychology, but quickly came to focus on psychology. He took philosophy courses with the new realists Holt and Perry, and psychology courses with Yerkes and Münsterberg. Tolman became interested in the behaviorist approach when Yerkes assigned Watson's 1914 text *Behavior* as required reading in his comparative psychology course. He spent the summer of 1912 in Germany, where he studied with Koffka. Although he never embraced Gestalt psychology, he remained interested in the Gestalt approach (he returned to Germany in 1923 to pursue his interest) and appropriated some of its concepts, such as the notion of a “sign-Gestalt.” Tolman completed his PhD in 1915 with a thesis supervised by Münsterberg on the learning of meaningless syllables conjoined with pleasant and unpleasant odors. *Studies in Memory* was published in *Psychological Monographs* in 1917.

Upon graduation from Harvard, Tolman took up an appointment at Northwestern University. He was dismissed in 1918, for either his poor teaching or his pacifism. He wrote an essay on the subject just after the United States entered the First World War, and it remained a strong personal theme throughout his life. He later wrote a book exploring the motivation for war, entitled *Drives Towards War* (1942), although his pacifism did not prevent him from serving in the Office of Strategic Services during the Second World War.

Tolman joined the University of California at Berkeley, where he taught comparative psychology and set up a laboratory for animal research. He began to work with rats and formed such a genuine affection for the animals that he dedicated his 1932 book to them. As Tolman put it,

Rats do not kill each other off in war; they do not invent engines of destruction . . . they do not go in for either class conflicts or race conflicts. . . . They are marvelous, pure and delightful.

—(1945, p. 166)

He remained at Berkeley for most of the rest of his career. He managed to avoid trouble with the university over his pacifism, but was suspended from 1950 to 1953 for refusing to take a loyalty oath (during which time he taught at Chicago and Harvard). Tolman took the university to court and won his case, and he later had the personal satisfaction of having the university acknowledge the moral validity of his position when he was awarded an honorary doctorate in 1959. He was elected president of the APA in 1937 and received its Distinguished Scientific Contribution Award in 1957. While Tolman may not have had as revolutionary an impact as Watson or as immediate an influence as Hull, his theoretical position has endured better than theirs. He claimed to have enjoyed his academic career from beginning to end. In commenting upon the guiding principles of his psychological research, he admitted that "In the end the only sure criterion is to have fun. And I have had fun" (1959, p. 159).

Purposive Behaviorism Tolman followed Watson in rejecting introspective psychology and maintaining that observable behavior is the subject matter of psychology, but he dismissed Watson's form of behaviorism as "muscle-twitchism." Tolman's own research was focused on purposive or goal-directed behavior in animals and humans. He called his version of behaviorism **purposive behaviorism** and titled his first book *Purposive Behavior in Animals and Men* (1932).

Like McDougall, who developed an earlier form of purposive behaviorism (McDougall, 1908), Tolman claimed that most animal and human behavior is intentionally directed toward goal or end states. As he put it, there is "something either *toward which* or *from which* the behavior is directed" (Tolman, 1925, p. 39). Like Aristotle, Tolman maintained that most animal and human behavior is intrinsically purposive or teleological. He insisted that the purposiveness or goal-directedness of behavior is "quite an objective and purely behaviorist affair":

It is a descriptive feature immanent in the character of behavior qua behavior. It is not a mentalistic entity supposed to exist parallel to, and to run alongside of the behavior. It is *out there* in the behavior, of its descriptive warp and woof.

—(1926, p. 355)

Tolman was influenced by the new realism of Holt and Perry, who maintained that consciousness and cognition should be explicated in terms of behavioral adjustments to the environment (Smith, 1986). Yet although he objected to inferences about mentality based upon introspective psychology (and to conceptions of purposive behavior as behavior accompanied by special introspective states), Tolman did not deny the independence of conscious mentality and behavior. He thought it was perfectly legitimate to postulate mental

determinants of the purposive behavior of animals and humans (as McDougall had before him):

For the behaviorist, “mental processes” are to be identified and defined in terms of the behaviors to which they lead. “Mental processes” are, for the behaviorist, naught but inferred determinants of behavior, which ultimately are deducible from behavior. Behavior and these inferred determinants are both objectively defined types of entity.

—(Tolman, 1932, p. 3)

Accordingly, Tolman postulated a variety of cognitive determinants of behavior, such as “internal presentations,” “representations,” “expectations,” “hypotheses,” “sign-Gestalts,” and “cognitive maps” and developed increasingly complex cognitive theories of animal learning in contrast to traditional stimulus-response accounts. He rejected the common conception of learning as the automatic connection of stimulus and response, based upon principles of contiguity, frequency, and reinforcement:

Behavior (except in the case of the simplest reflexes) is not governed by simple one-to-one stimulus response connections. It is governed rather by more or less complicated sets of patterns of adjustment which get set up within the organism.

—(Tolman, 1928, p. 526)

During his years at Berkeley, Tolman and his students developed a set of cognitive theories of learning based upon the experimental analysis of maze-learning behavior in rats. He claimed that animals learn “representations” rather than stimulus-response connections, often independently of motivation and reinforcement. On the basis of a famous series of experiments on **latent learning**, Tolman and Honzik (1930) maintained that rats are able to learn maze layouts in the absence of reinforcement. Tolman held that rats develop cognitive maps of their spatial environment based upon “confirmations” of their “expectancies”: “in the course of learning something like a field map of the environment gets established in the rat’s brain” (Tolman, 1948, p. 192).

Tolman’s commitment to the cognitive determination of both animal and human behavior was a distinctive feature of his purposive behaviorism. It distinguished it from the forms of behaviorism advocated by most other behaviorists, including Watson, Hull, and Skinner, who all denied that cognition and consciousness play a role in the determination of animal and human behavior. Tolman rejected Lloyd Morgan’s claim that all animal behavior could be explained in terms of association and never embraced the standard behaviorist attempt to accommodate all forms of animal and human behavior in terms of conditioned learning, based upon the experimental analysis of animal behavior. In his 1937 APA presidential address he claimed that much of “what is important in psychology” can be “investigated in essence through the continued experimental and theoretical

analysis of rat behavior at a choice point in a maze," but doubted whether this was true for human social, moral, and linguistic behavior or "such matters as involve society and words" (Tolman, 1938, p. 34). Also unlike most other behaviorists, including Watson, Hull, and Skinner, Tolman was not an environmentalist. Like McDougall, he maintained that a significant portion of animal and human behavior is innately determined (Tolman, 1936/1951).

Intervening Variables and Hypothetical Constructs Tolman developed his basic theoretical position in the 1920s (Tolman, 1922, 1923, 1925, 1926, 1927), but refined it in the language of scientific empiricism in the 1930s. Moritz Schlick, the founder of the Vienna Circle of logical positivists, visited Berkeley in 1931, and in 1933–1934 Tolman spent his sabbatical in Vienna, where he met with some of the original members of the circle. As a result of his exposure to logical positivism, Tolman came to treat cognitive states as intervening variables, operationally defined in terms of external environmental or internal physiological stimulus variables (independent variables) and behavioral response variables (dependent variables):

Mental processes, whether they be those of another or of ourselves, will figure only in the guise of objectively definable intervening variables. . . . the sole "cash-value" of mental processes lies, I shall assert, in their character as a set of intermediating functional processes which interconnect between the initiating causes of behavior, on the one hand, and the final resulting behavior, on the other. . . . Mental processes are but intervening variables between the five independent variables of (1) environmental stimuli, (2) physiological drive, (3) heredity, (4) previous training, and (5) maturity, on the one hand, and the final dependent variable, behavior, on the other.

—(1936/1951, pp. 116–117)

Tolman also called his form of purposive behaviorism **operational behaviorism**, because of his commitment to an "operational psychology" grounded in the principles of scientific empiricism. He claimed that such a psychology "seeks to define its concepts in such a manner that they can be stated and tested in terms of concrete operations by independent observers":

The behaviorism which I am going to present seeks, then, to use only concepts which are capable of such concrete operational verification.

—(1936/1951, p. 89)

Although Tolman championed the scientific empiricist account of theories and insisted upon the empirical grounding of theoretical postulates, he did not embrace the instrumentalist conception of psychological theories. He was a committed realist who treated intervening variables as theoretical references to internal cognitive

states that play an essential role in the causal explanation of animal and human behavior, and which are not reducible to mere descriptions of correlation between observable stimuli and responses.

The point may be put in terms of MacCorquodale and Meehl's (1948) later distinction between **hypothetical constructs** and intervening variables. They distinguished between intervening variables, for which "the statement of such a concept does not contain any words which are not reducible to the empirical laws," and hypothetical constructs, which are "not wholly reducible to the words in the empirical laws" (1948, p. 107). They claimed that hypothetical constructs, unlike intervening variables, contain **surplus meaning** that is not reducible to empirical laws. Tolman's theoretical cognitive postulates were hypothetical constructs rather than intervening variables, as he later acknowledged:

To use MacCorquodale and Meehl's distinction, I would now abandon what they call pure "intervening variables" for what they call "hypothetical constructs."

—(Tolman, 1949, p. 49)

However, Tolman did not restrict the surplus meaning of his theoretical cognitive constructs to an existential commitment to causally efficacious internal states of organisms, which MacCorquodale and Meehl granted was sufficient to distinguish a hypothetical construct from a pure intervening variable (according to this minimalist conception, a causally intervening variable counts as a hypothetical construct). Rather, he specified the meaning of his theoretical cognitive constructs *independently* of any operational definition in terms of observational stimuli and responses. Tolman specified his theoretical references to cognitive maps, for example, in terms of an organism's representation of its spatial environment, not in terms of its behavior in that environment. Given this theoretical specification, one could understand how rats represent their environment *without* knowing how they behave in that environment.

Clark L. Hull: A Newtonian Behavioral System

Hull came from a poor rural family in New York State. He worked on his parents' farm while attending local schools and survived a typhoid outbreak that took the lives of some of his schoolmates. He worked his way through Alma College, where he studied mining engineering. Hull often had to interrupt his school and college career to replenish his finances by taking on a variety of jobs, including teaching in his former high school, working as an apprentice mining engineer in Hibbing, Minnesota, and teaching at a normal school of education in Kentucky. A polio attack at the age of 24 ended his hopes of a mining career. Hull was left paralyzed in one leg and wore a heavy iron leg brace (which he designed himself)

for the rest of his life. He switched to psychology as a career that combined his joint interests in theory and apparatus, as well as promising speedy promotion (Hull, 1952). He read James's *Principles* while working as a high school teacher and continued his studies at the University of Michigan, graduating in 1913. Hull took courses in experimental psychology and logic and turned his engineering skills to the development of a machine for processing logical syllogisms. When he had saved enough money, he attended the University of Wisconsin, where he gained his PhD in 1918 at the age of 34. (Hull also applied to but was rejected by Yale, where he later spent the bulk of his professional career as a psychologist.) He was invited to join the faculty at the University of Wisconsin, where he remained as an instructor until 1929 (Hull, 1952).

Hull was proud of his dissertation on concept formation, supervised by Joseph Jastrow, which he considered to be a genuine experimental study of thought, and was deeply disappointed when it failed to generate much critical attention after being published in *Psychological Monographs* in 1920. Reprising Hume's comment on his *Treatise*, Hull described his first work as "still-born." However, as in the case of Hume, his comment was both exaggerated and premature. His work was regularly cited after J. P. Dashiell drew attention to it in his 1928 textbook *Fundamentals of Objective Psychology* (Hilgard, 1987).

During his years at Wisconsin, Hull conducted experimental studies of hypnosis and the effects of tobacco on efficiency. His critical survey of current theory and research on aptitude testing (Hull, 1928) cemented his growing reputation. While working in this area, Hull created a mechanical device for computing correlations, to avoid the time and effort of paper and pencil calculations. His "correlation machine" is now housed in the Smithsonian Museum in Washington. Hull was originally interested in Gestalt psychology and was instrumental in persuading the university to invite Koffka as a visiting professor. However, Hull was not persuaded by his advocacy of Gestalt principles. He later claimed that Koffka's critique of Watson's behaviorism inspired him to improve behaviorism rather than embrace Gestalt psychology (Hull, 1952).

In 1929 Hull accepted an appointment as research professor at Yale University, where he headed a group of researchers at the Institute of Human Relations. Throughout the following two decades Hull and his colleagues and students developed a program of research devoted to the study of the basic principles of learning, based upon maze-running experiments with rats. Hull also continued his research on hypnosis at Yale and published *Hypnosis and Suggestibility: An Experimental Approach* in 1933. However, the university directed Hull to discontinue his research after a female subject brought a lawsuit claiming that hypnosis was responsible for her nervous breakdown.

Hull admired Pavlov's work on conditioned reflexes, which he read in 1927 when it first appeared in English translation. He began research on conditioned responses at Wisconsin and continued with it at Yale. At the time it was generally

agreed that Watson and Pavlov had failed to integrate instrumental conditioning (trial-and-error learning) and classical conditioning (Boakes, 1984), and Tolman (1932) maintained that they were two distinct forms of learning. Hull developed a drive-reduction theory of reinforcement in which he attempted to accommodate both forms of conditioning under a single set of learning principles (by treating the unconditioned stimuli of classical conditioning as reinforcers):

Pavlov's conditioned reactions and the stimulus-response "bonds" resulting from Thorndike's so-called "law of effect" are in reality special cases of the operation of a single set of principles.

—(1937, p. 11)

Hull claimed that all learning is based upon the reduction of drives related to an organism's primary biological needs (such as food, drink, sexual gratification, temperature regulation, and relief from pain) and that reinforcement is equivalent to drive reduction or satisfaction. He theorized that the reaction potential of a response (sEr), the probability that a learned response will be emitted on any particular occasion, is a function of the habit strength (sHr) of the response, the number of times the response has been reinforced, and the level of drive (D). Hull extended his learning theory to accommodate complex behavior by appeal to secondary drives (and secondary reinforcement) that become associated (via classical conditioning) with the reduction of primary drives (primary reinforcement).

Hull developed his theoretical system in a series of papers in the 1930s, culminating in the publication of his *Principles of Behavior* in 1943. In this work, he formally presented his theory of learning as a set of theoretical axioms and derived empirical theorems. Hull modeled this presentation upon the system of Newton's *Principia*, a copy of which he had purchased while lecturing at Harvard in the summer of 1930; he required his graduate students to read and digest the work. Newton's gravitational theory was based upon a set of postulated axioms from which empirical laws about the motions of physical bodies were deduced, and Hull's theory of learning was based upon a set of postulated axioms from which empirical laws of observable behavior were deduced. Hull was not modest about the explanatory scope of his theory, based upon maze-running experiments on rats. In the *Preface* of his *Principles of Behavior*, he claimed that

all behavior, individual and social, moral and immoral, normal and psychopathic, is generated from the same primary laws.

—(1943a, p. v)

Unlike Tolman, Hull did not have fun. Throughout his life he was driven by the fear that he would die before completing his theoretical project, and his life ended in disappointment. He was forced to retract most of his earlier claims about

the generality of his theory in *A Behavior System*, published in 1952, the year he died from a heart attack. He regretted that he had never been able to complete a planned work that would have extended his theory to accommodate human moral and social behavior, and it was left to Hull's students, notably Robert Sears, John Dollard, and Neil Miller, to complete this part of his theoretical project (Dollard & Miller, 1950; Sears, 1943). Hull's theory was enormously influential in the 1940s (Spence, 1952), and his achievements in psychology were widely acknowledged. He was elected president of the APA in 1936, and in 1945 he was awarded the Warren Medal by the Society of Experimental Psychologists (the institutional descendant of Titchener's Experimentalists).

Intervening Variables and Cognitive Constructs Like Tolman, Hull endorsed the principles of scientific empiricism, which he claimed had the potential of transforming psychology into a "full-blown natural science" (1943b, p. 273). He claimed that theoretical intervening variables such as "drive" and "habit strength" must be related to observable independent and dependent variables:

It is evident that this equational mode of anchoring symbolic constructs to objectively observable and measurable antecedent and consequent conditions or phenomena is necessary, because otherwise their values would be indeterminate and the theory of which they constitute an essential part would be impossible of empirical verification.

—(Hull, 1943b, p. 282)

Also like Tolman, Hull rejected the instrumentalist conception of scientific theories. He treated theoretical postulates realistically, as causally intervening variables or hypothetical constructs. Although he avowed that theoretical references to habits must be operationally defined in terms of observable stimuli and responses, he maintained that habits exist independently of their particular behavioral expression in stimulus situations (as dispositions of the nervous system):

These symbols . . . represent entities or processes which, if existent, would account for certain events in the observable molar world. Examples of such postulated entities in the field of physical sciences are electrons, protons, positrons, etc. A closely parallel concept in the field of behavior familiar to everyone is that of *habit* as distinguished from habitual action. The habit presumably exists as an invisible condition of the nervous system quite as much when it is not mediating action as when habitual action is occurring; the habits upon which swimming is based are just as truly existent when a person is on the dance floor as when he is in the water.

—(1943a, p. 21)

Similarly, he claimed that theoretical references to drives must be defined in terms of observable stimuli and responses, but maintained that they exist independently

as internal states of an organism that causally mediate between environmental stimuli and behavioral responses.

Although he granted the existence of consciousness and cognition, Hull rejected most explanations of animal and human behavior in terms of them. He acknowledged that explanations in terms of consciousness are legitimate in principle, but held that there is no need to appeal to them in practice. In the abbreviated axiomatic theoretical system he presented in his 1936 APA presidential address (titled “Mind, Mechanism and Adaptive Behavior”), Hull claimed that theoretical references to consciousness are legitimate but unnecessary:

What, then, shall we say about consciousness? Is its existence denied? By no means. But to recognize the existence of a phenomenon is not the same thing as insisting on its basic, i.e., logical priority. Instead of furnishing a means for the selection of problems, consciousness appears to be itself a problem needing solution. In the miniature theoretical system, no mention of consciousness was made for the simple reason that no theorem has been found as yet whose deduction would be facilitated in any way by including such a postulate. . . . There is, however, no reason at all for not using consciousness or experience as a postulate in a scientific theoretical system if it clearly satisfies the deductive criteria already laid down.

—(1937, p. 30)

Hull doubted that theoretical references to consciousness would ever be needed to explain animal or human behavior:

Considering the practically complete failure of all this effort to yield even a small scientific system of adaptive or moral behavior in which consciousness finds a position of logical priority as a postulate, one may, perhaps, be pardoned for entertaining a certain amount of pessimism regarding such an eventuality.

—(1937, p. 31)

Hull also acknowledged the legitimacy of cognitive theoretical postulates and sometimes employed them in his own work (Hull, 1920, 1930), but he rejected Tolman’s theories of “representations” and “cognitive maps.” He believed that all behavior, including human moral and social behavior, could be explained in purely mechanistic terms. Hull and his colleagues and students at Yale tried to demonstrate that all apparently purposive behavior could be explained in terms of automatic principles of conditioning, without reference to cognitive states such as expectancies or cognitive maps. For example, Kenneth Spence maintained that the “systematic response patterns” that Tolman’s student Ivan Krechevsky had explained by reference to postulated “hypotheses” (Krechevsky, 1932) were simply products of reinforcement history (Spence, 1936). This generated a number of theoretical disputes between Hull and Tolman and their followers, including the famous debate

about whether rats running mazes learn cognitive maps or stimulus-response connections, the so-called **place versus response controversy** (Amundson, 1985).

There was also a fundamental difference between the theoretical cognitive constructs that Hull and Tolman employed. Hull agreed with Tolman that cognitive constructs are hypothetical constructs with surplus meaning that is not reducible to the mere description of correlations between observable stimuli and responses, but insisted that postulated cognitive states must be defined operationally as internal causal variables: as internal response-stimulus (r-s) connections that mediate between observable stimuli and responses. Tolman agreed that postulated cognitive states are internal causal variables that mediate between observable stimuli and responses and agreed on the need to relate cognitive constructs to observable stimuli and behavior. However, Tolman specified the meaning of his cognitive constructs (such as “cognitive map”) independently of their relation to observable stimuli and responses. In consequence, Hull’s avowedly “cognitive” theories lacked the autonomous cognitive content of Tolman’s theories.

This is worth noting in relation to recent claims that Hull anticipated contemporary forms of computational cognitive psychology through his discussion of the possibility of a **psychic machine** (Leahey, 1992; Schultz & Schultz, 1992). Hull conceived of a psychic machine (Hull, 1937, p. 30) as operating on automatic principles of learning based upon stimulus-response connections (including internal r-s connections), not as operating on the “rules and representations” postulated by contemporary computational cognitive psychologists (Bechtel, 1988). Hull also employed the notion of a psychic machine as a methodological device analogous to Morgan’s canon. He believed that the concept of a mechanized robot operating on automatic principles of conditioned learning served as a useful prophylactic against the theoretical attribution of cognitive states to animals, which helped to prevent such attribution from degenerating into “sheer anthropomorphism” (1943b, p. 287). He claimed that “one of the greatest obstacles to the attainment of a genuine theory of behavior is anthropomorphic subjectivism”:

One aid to the attainment of behavioral objectivity is to think in terms of the behavior of subhuman organisms, such as chimpanzees, monkeys, dogs, cats and albino rats. Unfortunately this form of prophylaxis against subjectivism all too often breaks down when a theorist begins thinking what he would do if he were a rat, a cat, or a chimpanzee; when that happens, all his knowledge of his own behavior, born of years of self-observation, at once begins to function in place of the objectively stated general rules or principles which are the proper substance of science.

A device much employed by the author has proved itself to be a far more effective prophylaxis. This is to regard, from time to time, the behaving organism as a completely self-maintaining robot, constructed of materials as unlike ourselves as may be.

—(1943a, p. 27)

The reference to “a theorist . . . thinking what he would do if he were a rat” was of course directed to Tolman. Like Watson before him, Hull was deeply suspicious of cognitive explanations of animal and human behavior.

While he acknowledged that cognitive states are internal states of organisms that play a causal role in the generation of behavior, Hull followed Watson in denying that they are “centrally initiated states.” He claimed that cognitive states are **pure stimulus acts**—defined as internal response-stimulus (r-s) sequences—that causally mediate between environmental stimuli and behavioral responses and which are fully determined by external stimuli (Hull, 1930). Like Watson, Hull was a committed environmentalist, albeit for different reasons.

Hull’s theories of conditioned learning had far greater initial impact than Tolman’s cognitive theories, although it is questionable whether this was a product of their superior theoretical fertility or empirical support. While the economic depression of the 1930s led to a general reduction in university funding, Hull’s program was the fortunate recipient of generous support from the Laura Spelman Rockefeller Memorial Fund. Beardsley Rumel (1894–1960), who had been a student of Angell’s at Chicago, was the administrator of the fund. Rumel had followed Angell to the Carnegie Corporation, where he served as Angell’s assistant prior to his appointment as administrator of the Rockefeller Fund. When Angell became president of Yale, he was highly successful in securing research grants from the Rockefeller Fund for the social sciences at Yale, including Hull’s program at the Institute of Human Relations. This meant reduced faculty teaching loads, better equipment, and support for graduate students, all of which enabled Hull’s program to thrive at a time when other universities (including Tolman’s University of California at Berkeley) were cutting back on their research budgets and hiring, while increasing the workload of their remaining faculty (Boakes, 1984).

Hull’s theories remained highly influential throughout the 1940s and 1950s. They were modified and extended by Kenneth Spence (1956, 1960) and his students at the University of Iowa and by former students and colleagues such as Carl Hovland (1912–1961) and O. Hobart Mowrer (1907–1982). Hull’s form of neobehaviorism remained the dominant force in American departments of psychology until the advent of radical behaviorism and the cognitive revolution in the 1950s and 1960s.

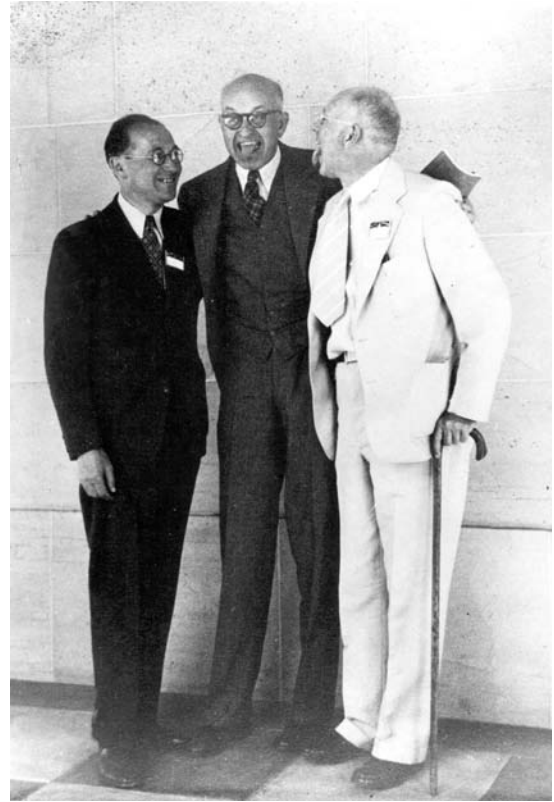
Neobehaviorist Theory and Operational Definition

The neobehaviorist conception of theoretical postulates as operationally defined intervening variables (instrumentally or realistically construed) impeded the development of psychological theory. For although neobehaviorists reasonably insisted upon the empirical grounding of psychological theories (the requirement that psychological theories generate testable predictions about observable behavior), their

commitment to the operational definition of theoretical meaning placed quite unnecessary restrictions on the content of psychological theories. The potentially fertile strategy of postulating internal states to explain observable behavior was transformed into a set of emaciating dogmas about the exhaustive operational definition of postulates in terms of observable stimuli and responses—a doubtful ideal that few neobehaviorists actually satisfied in practice (Estes et al., 1954).

What Is Learned? Nowhere is this more apparent than in Howard Kendler's (1952) supposed solution to the theoretical dispute between Hull and Tolman about "what is learned" by rats in mazes. Tolman claimed that rats develop a cognitive representation of the spatial layout of the maze (a cognitive map), whereas Hull claimed that rats learn differentially reinforced stimulus-response connections. Although both Tolman and Hull were committed realists who conceived of intervening variables as causally efficacious internal states, Kendler claimed that the debate about what is learned was a "pseudo issue," since theoretical postulates in psychology are nothing more than economical summaries or "'shorthand' descriptions" of the observable stimulus-response sequences in terms of which they are operationally defined. Given that both theories were operationally defined in terms of (and purported to explain) the same observable behavior in the same stimulus situations, Kendler maintained there was no difference in the content of the two theories—they amounted to the same theory. He held that to suppose otherwise, to treat theoretical postulates as references to independently real entities or processes, was to commit "the fallacy of reification or hypostatization":

The construct of learning, whether it be conceived in terms of modifications in cognitive maps or S-R connections, does not refer to an object, thing, or entity as is suggested by those who are concerned with the question of what is learned. These intervening variables possess no meaning over and above their stated relationships between the independent and dependent variables. The basic error underlying the problem of what is learned is the assumption that these intervening variables are



The tongues of men: Lewin, Tolman, and Hull, all having fun.

entities capable of being described and elaborated on, independently of their operational meaning.

—(1952, pp. 271–272)

If Kelman had been correct that Tolman's and Hull's theories were wholly specified in terms of the same relations between observable independent and dependent variables, this would have precluded the empirical development and evaluation of these theories, as it would with respect to any competing theories that shared the same empirical domain. The absurdity of this position is perhaps best illustrated by considering the equivalent suggestion to the effect that there was nothing at issue between the Copernican and Ptolemaic astronomical theories, or the wave and particle theories of light, just because they shared the same empirical domain (the same empirical predictions could be derived from both sets of theories).

Theoretical Meaning and Operational Measures The neobehaviorist error, inherited from scientific empiricist philosophy of science, was to confuse the reasonable demand for operational measures of postulated theoretical states and processes with the peculiar notion that the meaning of theoretical postulates must be specified in terms of such operational measures. The meaning of theoretical postulates in natural science is generally determined independently of operational definition. This explains why it is possible to understand Bohr's theory of the atom, for example, which postulates that electrons are held in orbit by a nucleus of protons and neutrons, without having any inkling of how changes in the energy level of electrons explain differences in the observed spectral emission lines for elements. For the same reason it is possible to understand Tolman's theoretical descriptions of cognitive maps without knowing how rats behave in mazes.

Of course, operational measures of theoretical constructs such as "electron" and "short-term memory" are important, but they do not determine their meaning, as evidenced by the fact that their meaning remains invariant when different operational measures are employed in different empirical or experimental contexts. Theoretical claims about electrons, for example, do not change their meaning between experiments in which the presence of electrons is demonstrated through spectography or tracks in a Wilson cloud chamber. Similarly, theoretical claims about short-term memory do not change their meaning when they are evaluated by experimental studies employing verbal as opposed to written measures.

The substantive differences in the contents of competing theories in natural and psychological science derive from their independent contents, which are not defined operationally. The substantive differences between the Copernican and Ptolemaic theories, and the wave and particle theories of light, were differences in "surplus" meaning. These generated the different predictions (about the stellar parallax and the speed of light in air and water) that eventually enabled disputes

between these theories to be empirically adjudicated. The inconclusiveness of the original Tolman-Hull debate about “what is learned” is probably best explained in terms of the temporary ability of both theories to accommodate failed predictions via auxiliary hypothesis modification, which is the best explanation for the long periods of time during which the debates between the Copernican and Ptolemaic theories and the wave and particle theories of light were empirically undecidable, as Kendler (1981) later acknowledged.

Yet most neobehaviorists would have none of this. They insisted that surplus meaning has no place in psychological science and maintained that “operationally valid intervening variables . . . are the only kinds of constructs ultimately admissible in sound scientific theory” (Marx, 1951, p. 246). This placed quite unnecessary constraints on the development of psychological theory and created a critical conceptual dilemma for neobehaviorists. If the meaningful content of theoretical postulates really is exhaustively determined by the observational stimulus-response sequences in terms of which they are operationally defined, then why not simply dispense with theoretical constructs altogether and restrict psychological science to the description of observational stimulus-response sequences? As one scientific empiricist put it,

If the terms and principles of a theory serve their purpose, that is, they establish definitive connections among observational phenomena, then they can be dispensed with, since any chain of laws and interpretative statements establishing such a connection should then be replaceable by a law that directly links observational antecedents to observational consequents.

—(Hempel, 1965, p. 186)

The radical behaviorist B. F. Skinner grasped the nettle of this implication, often characterized as the **theoretician’s dilemma**, and maintained that putative theoretical references to internal cognitive states are circular and unnecessary “explanatory fictions” (Skinner, 1953).

RADICAL BEHAVIORISM

Burrhus F. Skinner was born in 1904 in Susquehanna, Pennsylvania, into a middle-class family (his younger brother died at an early age). At high school, Skinner did poorly in science, but excelled in literature. He attended Hamilton College in Clinton, New York, where he gained a reputation as a trickster and prankster. After graduating with a BA in English literature, Skinner tried his hand as a writer. He lived in Paris and Greenwich Village and smoked a pipe, but he frittered his time away and produced nothing (although Robert Frost did comment favorably on some of his work at a summer writing school). He became depressed and considered himself a failure in love as well as in literature, having been rejected by half

a dozen girls who took his fancy. Unlike many earlier psychologists, this disillusioned writer was not inspired by James's *Principles*, but by the work of Watson and Pavlov, which stimulated him to follow a career in psychology.

Skinner was accepted into the graduate program in psychology at Harvard in 1928 (where he eventually spent most of his career). He devoted himself to psychology and physiology during his years at Harvard, where he was influenced by Walter S. Hunter (1889–1954), who introduced Skinner to behaviorism in a course he taught on animal behavior (Skinner, 1967). Skinner heard Pavlov lecture at the 1929 International Congress in Psychology held at Harvard and obtained a signed photograph of his hero (which he hung over his desk). He earned his master's degree in 1930 and his PhD in 1931, and he continued at Harvard for five more years as a postgraduate fellow (of the Harvard Society of Fellows). He taught at the University of Minnesota from 1936 to 1945, during which time he published *The Behavior of Organisms* (1938). The book received mixed reviews and had little initial impact (the first print run of 800 copies took about 10 years to sell), being overshadowed by the anticipated publication of Hull's *Principles of Behavior*. Appleton-Century-Crofts was initially reluctant to publish the book, having already contracted for Hull's *Principles*, but was persuaded by an enthusiastic external review by Tolman and a publication grant from Harvard.

Operant Conditioning

Skinner's research focused upon what he called operant behavior, as opposed to Pavlovian respondent behavior. He defined **operant behavior** as behavior whose probability of recurrence is increased by reinforcement: an "emitted" behavior is positively or negatively reinforced via a food reward or removal of a painful stimulus. In contrast, he defined **respondent behavior** as behavior "elicited" by unconditioned and conditioned stimuli. According to Skinner, **operant conditioning** increases the probability of the recurrence of a behavior (in a stimulus situation) and the frequency of its recurrence, through reinforcement of the behavior. Thus a rat in a Skinner box (a box specially constructed to eliminate potentially interfering stimuli and to control the reinforcement of emitted behavior) will eventually depress a lever that delivers a food pellet into a tray, and the rat's rate of bar pressing will increase cumulatively once it begins to be reinforced. Skinner's *Behavior of Organisms* (1938) described the basic principles of operant conditioning, including reinforcement, extinction, spontaneous recovery, discrimination learning, and the effect of punishment on learning. This work demonstrated Skinner's ability to predict and control the behavior of laboratory rats and established his reputation as a rigorous experimentalist.

Skinner moved to Indiana University in 1945, where he served as chair of the department of psychology until 1948. He then returned to Harvard, where he

remained (as professor emeritus from 1974) until his death in 1990 (at the age of 86 from leukemia). He published *Science and Human Behavior* in 1953, *Verbal Behavior* in 1957, and *About Behaviorism* in 1974. From the 1940s onward Skinner began to explore the influence of different **schedules of reinforcement** on learning, such as fixed and variable interval and ratio schedules (Ferster & Skinner, 1957), which he considered to be his most significant contribution to scientific psychology. He also investigated the **shaping** of behavior through the reinforcement of progressive approximations to a target behavior (Skinner, 1951), such as the pecking of piano keys by pigeons and vocalization by catatonic schizophrenics.

Skinner claimed that his account of operant conditioning was merely a refinement of Thorndike's "law of effect." However, Skinner abjured Thorndike's references to satisfaction and discomfort and rejected Hull's interpretation of reinforcement as drive reduction. He maintained that "the only defining characteristic of a reinforcing stimulus is that it reinforces" (1953, p. 72), ignoring the obvious circularity of the definition.

Explanatory Fictions

Skinner rejected the scientific empiricist and neobehaviorist account of psychological theories. He claimed that theories about unobservable states and processes, including theories about cognitive states and processes, are **explanatory fictions**: They are vacuous as explanations of relations between observable stimuli and responses and play no role in the development of novel predictions about behavior. According to the neobehaviorist account of the meaning of psychological theories, intervening variables must be operationally defined in terms of observable stimuli and responses. Skinner argued that if theoretical postulates such as "cognitive maps" or "habit strength" really were defined in terms of empirical laws relating observable stimuli and responses, then any avowed explanation of empirical laws in terms of "cognitive maps" or "habit strength" would be vacuously circular. According to Skinner, such an "explanation" would appeal to the very functional relationships that it purported to explain:

To what extent is it helpful to be told, "He drinks because he is thirsty"? If to be thirsty means nothing more than to have a tendency to drink, this is mere redundancy.

—(1953, p. 33)

For similar reasons, he maintained that there is no need to appeal to intervening variables in the "practical control of behavior":

When an example of maladjusted behavior is explained by saying that the individual is "suffering from anxiety," we still have to be told the cause of the

anxiety. But the external conditions which are then invoked could have been directly related to the maladjusted behavior. Again, when we are told that a man stole a loaf of bread because “he was hungry,” we still have to learn of the external conditions responsible for the “hunger.” These conditions would have sufficed to explain the theft.

—(1953, p. 35)

Skinner also recognized that the notion that operationally defined intervening variables can generate novel empirical predictions is completely illusory. Since at any point in time, the content of postulated intervening variables is supposedly determined by the functional relationships between observable stimuli and responses in terms of which they are operationally defined, the only basis for the prediction of novel functional relationships *is our present knowledge of functional relationships between observable stimuli and responses*. Skinner claimed that the whole apparatus of intervening variables and operational definitions is redundant with respect to the explanation and prediction of observable behavior, which is ultimately based upon functional laws relating observable stimuli and behavior:

The objection to inner states is not that they do not exist, but that they are not relevant in a functional analysis. We cannot account for the behavior of any system while staying wholly inside it; eventually we must turn to forces operating on the organism from without. Unless there is a weak spot in our causal chain so that the second link is not causally determined by the first, or the third by the second, then the first and third links must be lawfully related. If we must always go back beyond the second link for prediction and control, we may avoid many tiresome and exhausting digressions by examining the third link as a function of the first.

—(1953, p. 35)

These were legitimate criticisms of the standard neobehaviorist account of theoretical meaning in terms of operationally defined intervening variables, but they did not extend to the “surplus meaning” of Tolman’s theoretical constructs or those developed by later cognitive theorists.

Radical Behaviorism

Skinner’s radical behaviorism marked a return to the positivist and inductivist form of behaviorism championed by Watson, with an emphasis on response reinforcement contingencies rather than stimulus-response connections.

Skinner maintained that behaviorist psychology should focus on the description of functional relationships between observable behavior and environmental

stimuli, based upon the experimental analysis, control, and manipulation of environmental reinforcement:

We can predict and control behavior, we can modify it, we can construct it according to specifications, and all without answering the explanatory questions which have driven investigators into the study of the inner man.

—(Skinner, 1961, p. 254)

Skinner eschewed theory and rejected the hypothetico-deductive method advocated by Tolman and Hull. In “Are Theories of Learning Necessary?” (1950), he argued that significant experimentation in psychology does not need to be theory-guided, but can proceed via the inductive accumulation of functional laws relating observable stimuli and responses, or behavior and reinforcement.

Skinner characterized his behaviorism as **descriptive behaviorism** for the same positivist reasons that Titchener characterized his structural psychology as descriptive psychology. Skinner and Titchener rejected explanatory appeals to unobservable states and processes and equated causal explanation with the description of observable correlation. However, Skinner’s frequent presentation of his radical behaviorist position as atheoretical was somewhat disingenuous, since his own research program was firmly grounded upon distinctive theoretical commitments, as he later admitted (Skinner, 1967).

Like Watson and Hull, Skinner acknowledged the existence of cognitive states, but denied that they are “centrally initiated.” Like Hull (and Sechenov), Skinner maintained that cognitive states are merely internal links in fully determined causal chains relating behavior and environmental stimuli. He rejected the notion that the cognitive agent is the “true originator or initiator of action” (Skinner, 1974, p. 225) and, like most other behaviorists, was a committed environmentalist.

Skinner’s theoretical commitment to the environmental determination of behavior led him to champion programs of social engineering based upon experimentally derived principles of behavior modification. Skinnerian principles of behavior modification came to be employed in a wide variety of industrial, educational, and clinical training programs, as well as commercial animal training services (such as Animal Behavior Enterprises, formed by Keller and Marian Breland, two former students of Skinner’s). Managers employed reinforcement schedules to maximize production in factories, educators introduced teaching machines to schools and colleges, and clinicians employed token-economy programs to shape the behavior of patients in psychiatric hospitals (in which patients could trade tokens for reinforcers such as candy or cigarettes). Clinical psychologists adopted forms of behavior therapy based upon Skinnerian principles of behavior modification, which remain popular to this day.

Skinner’s reprisal of Watson’s entry into the realm of child rearing was less successful and brought him some notoriety. He designed an air-crib that allowed



Baby in a box: Yvonne and Debby Skinner.

parents to manipulate an infant's environment, based upon the same principles as the Skinner box. His eldest daughter Deborah was raised in this "baby-box" for the first two and a half years of her life. The device was described in an article titled "Baby In A Box," published in *Ladies' Home Journal* in 1945 and received extensive coverage in the media. While it aroused interest among some parents seeking relief from the more mundane tasks of child rearing (such as laundry—the crib was temperature controlled, so there was no need for heavy clothing or blankets), it also generated outrage from those who considered it a cruel practice, analogous to raising children like household pets. About 130 parents used Skinner's "air-crib," or "heir-conditioner," as he called it.

Rumors later abounded about how Deborah Skinner had been psychologically scarred by her isolation in the baby box and (like Watson's children) had become psychotic and suicidal in later life. These rumors proved to be mere wishful thinking on the part of Skinner's critics. Deborah had happy memories of her childhood, and after graduating with honors from Radcliffe College, embarked on a successful artistic career. Skinner's other daughter Julie was not raised in an "air-crib" but raised her own children in one.

In 1948 Skinner published *Walden Two*, a novel about a community run on principles of behavior modification. The book was a bizarre cross between Aldous

Huxley's *Brave New World* and Henry David Thoreau's utopian *Walden*. In 1971 Skinner published *Beyond Freedom and Dignity*, in which he dismissed the notion of human freedom as a superstition that impeded the application of the methods of science to the control of human behavior. The book became a best seller and made Skinner a household name. In television shows and newspaper articles, Skinner (like Watson before him) became something of an academic celebrity and (also like Watson before him) relished the controversy generated by his provocative remarks. When asked whether, if he had to choose, he would burn his books or his children, Skinner picked the latter, a response that generated predictable public outrage.

In the decades after the Second World War, Skinner's radical behaviorism became a distinctive movement within scientific psychology. The *Journal for the Experimental Analysis of Behavior* was founded in 1958 as a publication outlet for radical behaviorists (to avoid the statistical publication conditions of other psychology journals that Skinner and his followers rejected) and the *Journal of Applied Behavior Analysis* in 1968 (to cope with the increased volume of radical behaviorist submissions). Division 25 of the American Psychological Association was designated as the Division of the Experimental Analysis of Behavior. Radical behaviorists came to play an influential role in the development of many psychology departments, notably at Columbia University, where Fred Keller, a friend and fellow-student of Skinner's at Harvard, and his student William Nathan (Nat) Schoenfeld transformed the curriculum in accordance with radical behaviorist principles (Bolles, 1993). This was perhaps apt, since it was at Columbia that the "behaviorist revolution" began with Watson's 1913 lecture.

Skinner was the most famous 20th-century psychologist, both for his substantive scientific achievements and for his controversial public persona. He was awarded the Warren Medal by the Society of Experimental Psychologists in 1942, a Distinguished Scientific Contribution Award by the APA in 1958, and the Gold Medal of the American Psychological Foundation in 1971. He was never elected president of the APA, but served as president of the Midwest Psychological Association in 1948.

Skinner had no truck with the later cognitive revolution in psychology. He believed that it betrayed the principles of scientific psychology and marked a return to superstition and introspection. He accused cognitive psychologists of employing unscientific speculations about "internal processes" as the causes of behavior and of "emasculating the experimental analysis of behavior" (1985, p. 300).

Skinner remained active intellectually until his death in 1990. In 1983 (at the age of 78) he published *Enjoying Old Age* (with Margaret Vaughan), a behaviorist primer for remaining active and happy in old age, based upon his own practical experience. In 1990 he received a Lifetime Contribution to Psychology Award from the APA, in recognition of his "dynamic and far-reaching-impact on the discipline" (1990, p. 1205). Skinner accepted the Award Citation at the 1990 meeting of the APA in Boston, where he made his final public appearance, delivering a

critical address titled “Can Psychology Be a Science of the Mind?” (published in *American Psychologist*, November 1990). He died a week later.

THE SECOND WORLD WAR AND THE PROFESSIONALIZATION OF ACADEMIC PSYCHOLOGY

Skinner’s form of radical behaviorism displaced neobehaviorism in the decades following the Second World War, although it was later overshadowed by the cognitive revolution in psychology, which was stimulated by theoretical and technical developments during the war. The Second World War, which began in Europe when Germany invaded Poland in 1939 and which America entered after the Japanese bombed Pearl Harbor in 1941, also precipitated a major transformation of the structure of scientific and professional psychology.

The heady public optimism about the potential of psychology in the 1920s was replaced by general public skepticism in the 1930s, as economic prosperity gave way to economic depression. Newspaper and magazine coverage of psychology declined. The general disillusionment with the promised social contribution of psychology was aptly expressed in a *New York Times* editorial in 1934, which lamented the fact that psychology was strong on promises during the good times but weak on the delivery of solutions to real problems in times of hardship (Benjamin, 1986).

Psychological Contributions to the War Effort

Once again, the contribution psychologists made to the war effort did wonders for their public reputation. In contrast to the First World War, psychologists were well prepared for the Second World War. The APA and the American Association of Applied Psychologists (AAP) organized planning committees prior to the outbreak of war. These committees held their first joint meeting in 1939 and merged in 1940 under government pressure for the integration of psychological services. Walter V. Bingham and Robert M. Yerkes, both veterans of psychological service in the First World War, worked vigorously to promote the military applications of psychology. Although they failed to persuade the army to adopt their comprehensive plan, American psychologists made significant contributions to the war effort, working through the office of the Emergency Committee of Psychology, which comprised representatives from the APA and AAP, the Society for the Psychological Study of Social Issues (SPSSI), the Society of Experimental Psychologists (SEP), and Section I (Psychology) of the American Association for the Advancement of Science (AAAS).

As in the First World War, much of the psychological war effort was focused on personnel selection and aptitude testing, which was directed by Bingham, who

became chairman of the Committee on Classification of Military Personnel (with the rank of colonel). However, psychologists also contributed to intelligence work and the evaluation of prisoners of war (Capshe, 1999). They worked with other social scientists and psychiatrists on studies of the attitudes, morale, and adjustment of combat troops (Stouffer, Lumsdane, et al., 1949; Stouffer, Suchman, et al., 1949), the psychological effects of saturation bombing on the enemy (U.S. Strategic Bombing Survey, 1946), French civilian reaction to the D-Day landings (Riley, 1947), and civilian morale and propaganda (Watson, 1942). Psychologists also played a major role in **human factors research**, which focused on the interaction between human operators and machines such as torpedoes and anti-aircraft guns, and radar and communication systems. This research promoted developments in information theory that stimulated the later cognitive revolution in psychology. Psychologists also exploited group dynamics to encourage wartime changes in food habits and the purchase of U.S. war bonds (Cartwright, 1949; Lewin, 1947), and they played a major role in the provision of psychotherapy and counseling for emotionally disturbed soldiers, since psychiatrists were overwhelmed by the sheer number of psychological casualties during the war (Benjamin et al., 2003).

The Reorganization of the APA

Government pressure for cooperation between scientific and professional psychologists during wartime led to the formation of the Office of Psychological Personnel (OPP), located in Washington, DC. Yerkes exploited the situation to try to repair the prewar split between the APA and the AAAP and organized a constitutional convention to plan for the postwar reconstruction of the APA. The American Psychological Association was reorganized and reconstituted in 1944 with 18 charter divisions, which represented the interests of applied psychologists such as clinical, consulting, industrial, and school psychologists, and the various scientific societies (such as the Society for the Psychological Study of Social Issues) and interest groups (such as the National Council of Women Psychologists). The divisional structure of the new APA was modeled upon the divisional structure of the AAAP, which disbanded that same year. According to the constitution of the new APA, the goal of the association was “to advance psychology as science, as a profession, and as a means of promoting human welfare” (Wolfe, 1946/1997, p. 721), in contrast to the goal of the original APA, which was restricted to “the advancement of psychology as a science.” Applied psychologists demanded the reference to the “profession” of psychology, and the reference to “promoting human welfare” was added at the last minute at the insistence of the Society for the Psychological Study of Social Issues. The Executive Secretariat of the new APA was housed in the former offices of the OPP in Washington, and *American Psychologist* was instituted in 1946 as the new “professional journal” of the APA (Benjamin et al., 2003).

Postwar Expansion

Clinical, consulting, and industrial psychology (three of the four original sections of the AAP) expanded dramatically after the war. Recognizing that there were too few psychiatrists to cope with the psychological needs of returning veterans, the government authorized the U.S. Public Health Service (USPHS) and the Veterans Administration (VA) to expand the pool of mental health professionals by creating and funding training programs in clinical psychology at major universities. The USPHS and the VA worked with the new APA to develop program evaluations, which led to accreditation programs in clinical psychology in 1946 and in counseling psychology in 1952 (Benjamin et al., 2003).

Industrial psychology also got a major boost from its perceived record of accomplishment in the war. Businesses increased their employment of psychological tests from around 14 percent in the prewar period to around 75 percent by 1950 (Napoli, 1981, p. 138, cited in Benjamin et al., 2003). As applied branches of psychology established their professional identity through the creation of specially tailored doctoral programs, so too did subdisciplines such as social and developmental psychology. Psychology became a popular subject once again, with newspapers, magazines (such as *Psychology Today*, founded in 1967), and a veritable avalanche of self-help books fueling the public demand for psychological advice and knowledge.

Neobehaviorists such as Carl Hovland contributed to the war effort through the study of mass communication (Hovland, Lumsdaine, & Sheffield, 1949), as did radical behaviorists such as Skinner and his students. Skinner worked with Keller and Marian Breland on Project Orcon, for which they trained pigeons as “missile guidance systems” or “organic control systems” (Skinner, 1960). The pigeons performed successfully in missile guidance simulations, but the military declined to adopt them. After the war, behaviorist theories of conditioned learning faced increasing problems, both theoretical and empirical, and were rejected by proponents of the emerging cognitive revolution in psychology. Radical behaviorists remained an active force in psychology, but became increasingly isolated in the fashion of Titchener’s structural psychologists at the beginning of the 20th century (Krantz, 1972).

PROBLEMS OF BEHAVIORISM

By the end of the Second World War, many psychologists had come to question the behaviorist presumption that all or most of human psychology and behavior could be explained in terms of general theories of conditioning based upon the experimental study of rats and pigeons. They doubted that such theories could be extended to accommodate human moral, social, and linguistic behavior, as most behaviorists had maintained (with the notable exception of Tolman). Neobehaviorists

themselves were forced to develop **mediational theories** that appealed to complex internal response-stimulus (r-s) sequences in order to accommodate linguistic behavior and symbolic meaning (N. Miller, 1959; Osgood, 1957).

In the 1950s and 1960s behaviorist theories based upon conditioning came under increasing attack, in terms of both the scope and basic principles of conditioning theory. Although none were perhaps sufficient to undermine the entire behaviorist program, these critiques prepared the way for the cognitive revolution in psychology.

Chomsky's Critique of Skinner

B. F. Skinner advanced the most detailed attempt to develop a behaviorist account of linguistic behavior in his 1957 book *Verbal Behavior*. In this work Skinner identified the referents of words (or, in his terminology, verbal operant responses, or "tacts") with discriminative stimuli in the environment that come to control the correct emission of verbal responses through reinforcement by the linguistic community. Skinner maintained that

The basic processes and relations which give verbal behavior its special characteristics are now fairly well understood. . . . most of the experimental work responsible for the advance of the experimental analysis of behavior has been carried out on other species. . . . the results have proved to be surprisingly free of species restrictions. . . . recent work has shown that the methods can be extended to human behavior without serious modification.

—(1957, p. 3)

Noam Chomsky (1928–) subjected Skinner's claim that the complexities of verbal behavior could be explained in terms of discriminative stimuli and reinforcement history to a devastating critique. In a famous review of Skinner's book in *Language* in 1959, Chomsky argued that Skinner's basic concepts of stimulus, response, and reinforcement could be objectively defined only in terms of rigorously controlled experiments and could not be extended to accommodate complex verbal behavior subject to "ill-defined factors of attention, set, volition and caprice" (1959, p. 30). He claimed that Skinner had failed miserably in his attempt to explain verbal behavior by reference to a "few external factors . . . isolated experimentally with lower organisms," without any appeal to the cognitive contribution of the speaker. Chomsky also suggested that

The magnitude of the failure of this attempt to account for verbal behavior serves as a kind of measure of the importance of the factors omitted from consideration, and an indication of how little is actually known about this remarkably complex phenomenon.

—(1959, p. 28)

Anticipating the form of explanation that would later become characteristic of cognitive psychology, Chomsky argued that human verbal behavior could only be explained in terms of representations of the rules governing the construction of sentences. According to Chomsky, our ability to recognize sentences cannot be explained in terms of behaviorist principles such as “stimulus generalization” or in terms of basic associationist principles (such as contiguity, frequency, and reinforcement), but must be explained in terms of the internal representation of the grammar of a language:

We constantly read and hear new sequences of words, recognize them as sentences, and understand them. It is easy to show that the new events that we accept and understand as sentences are not related to those with which we are familiar by any simple notion of formal (or semantic or statistical) similarity or identity of grammatical frame. Talk of generalization in this case is entirely pointless and empty. It appears that we recognize a new item as a sentence not because it matches some familiar item in any simple way, but because it is generated by the grammar that each individual has somehow and in some form internalized. And we understand a new sentence, in part, because we are somehow capable of determining the process by which this sentence is derived in this grammar.

—(1959, p. 56)

Chomsky stressed the extraordinary nature of the child’s achievement in learning a language:

The child who learns a language has in some sense constructed a grammar for himself on the basis of his observation of sentences and nonsentences (i.e. corrections by the linguistic community). Study of the actual observed ability of a speaker to distinguish sentences from non-sentences, detect ambiguities, etc., apparently forces us to the conclusion that this grammar is of an extremely complex and abstract character, and that the young child has succeeded in carrying out what from the formal point of view, at least, seems to be a remarkable type of theory construction. Furthermore, this task is accomplished in an astonishingly short time, to a large extent independently of intelligence, and in a comparable way by all children. Any theory of learning must cope with these facts.

—(1959, p. 57)

These considerations led Chomsky to doubt that any behaviorist theory could account for the remarkable fact that all normal children acquire “essentially comparable grammars with remarkable rapidity” (1959, p. 57), based upon a very limited set of data (the so-called “poverty of the stimulus” argument). According to Chomsky, the child’s ability to acquire grammar at such a rapid

rate can be explained only by the postulation of some innate “‘hypothesis-formulating’ ability” (1959, p. 57), a suggestion Chomsky later developed in his theory of (innate) transformational grammar—a postulated system of rules governing the production of grammatically well-formed sentences (Chomsky, 1965, 1966, 1972).

Although linguistic phenomena such as reference might appear to be susceptible to a behaviorist analysis in terms of stimulus control, Chomsky maintained that no behaviorist theory could explain the *structure* of grammar or the productivity and creativity of language. And Chomsky’s followers argued that not even reference could be explained in terms of stimulus control, since one of the distinctive features of language is that it is not “stimulus-bound”:

A striking feature of linguistic behavior is its freedom from the control of specifiable local stimuli or independently identifiable drive states. In typical situations, what is said may have no obvious correlation with conditions in the immediate locality of the speaker or with his recent history of deprivation or reward.

—(Fodor, 1965, p. 73)

Chomsky’s own linguistic theory underwent a multitude of transformations from *Aspects of a Theory of Syntax* in 1965 to *Language and the Problem of Knowledge* in 1988 and shaped the development of postwar linguistics to such a degree that it is not unreasonable to maintain that “the history of modern linguistics is the history of Chomsky’s ideas and of the diverse reactions to them on the part of the community” (Gardner, 1985, p. 185, original emphasis). Although Chomsky influenced many individual psychologists and persuaded some (such as George Miller) to abandon behaviorism and embrace cognitive forms of psychological explanation, his own distinctive theoretical position had little direct impact upon the development of later theories in cognitive psychology. While all cognitive psychologists embraced Chomsky’s **cognitivism**, his commitment to the legitimacy and utility of cognitive theoretical constructs, many were (and remain) reluctant to embrace the extreme nativism and formalism of his linguistic theory.

The Misbehavior of Organisms

While Chomsky’s critique was powerful, it demonstrated the limits of behaviorist explanation with respect to only one distinctively human behavior—language. Other critiques pointed to the limits of behaviorist explanation with respect to animal behavior, the original foundation of the behaviorist program. Behaviorists had generally depreciated the role of instinct in animal and human behavior. European biologists and ethologists such as Konrad Lorenz (1903–1989) and Nikolaas (Niko) Tinbergen (1907–1988), who received the Nobel Prize in physiology and

medicine in 1973 for their naturalistic studies of species-specific instinctual behavior (Lorenz, 1950; Tinbergen, 1951), criticized American “ratrunners” for their failure to study animals in their natural environment (ignoring Watson’s early field studies of the behavior of noddy and sooty terns) and their neglect of instinctual constraints on learning—criticisms that were echoed by American psychologists such as Frank Beach (1950, 1960).

Psychologists within the behaviorist camp also developed similar criticisms. Keller and Marian Breland were former students and colleagues of Skinner’s who founded a company called Animal Behavior Enterprises. They employed operant conditioning principles in training animals commercially for entertainment parks and television, but began to question some of the basic assumptions of conditioning theory. Their extensive practical experience in training animals taught them that it is extremely difficult for animals to learn certain forms of behavior: What they were eventually forced to recognize as instinctual behavior interfered with learning and came to displace learned behavior. The Brelands described numerous cases of animals that had been conditioned to a behavior but would then engage in a quite different behavior. They claimed that “it can easily be seen that these particular behaviors to which the animals drift are clear-cut examples of instinctive behaviors having to do with the natural food getting behaviors of the particular species” (1961, p. 683).

For example, although one could train raccoons and pigs to deposit wooden tokens into containers, they quickly reverted to instinctual “washing” and “rooting” behaviors. The Brelands described their attempt to teach a raccoon to drop a wooden “coin” into a “piggy bank”:

The raccoon really had problems (and so did we). Not only could he not let go of the coins, but he spent seconds, even minutes, rubbing them together (in a most miserly fashion), and dipping them into the container. He carried on this behavior to such an extent that the practical application we had in mind—a display featuring a raccoon putting money in a piggy bank—simply was not feasible. The rubbing behavior became worse and worse as time went on, in spite of nonreinforcement.

—(1961, p. 682)

They claimed that the raccoon was demonstrating the form of “washing behavior” that it would normally employ in the removal of the exoskeleton of a crayfish, for example, and concluded that

It seems obvious that these animals are trapped by strong instinctive behaviors, and clearly we have here a demonstration of the prepotency of such behavior patterns over those which have been conditioned.

—(1961, p. 684)

The Brelands called this phenomenon **instinctual drift**:

The general principle seems to be that wherever an animal has strong instinctive behaviors in the area of the conditioned response, after continued running the organism will drift toward the instinctive behavior to the detriment of the conditioned behavior and even to the delay or preclusion of the reinforcement.

—(1961, p. 684)

The Brelands presented a strong empirical case that there are significant evolutionary biological limits to animal learning. They claimed that their examples of instinctual drift demonstrated the “clear and utter failure of conditioning theory” (1961, p. 683) and maintained that animal behavior “cannot be adequately understood, predicted or controlled without knowledge of its instinctive patterns, evolutionary history, and ecological niche” (1961, p. 684). In a similar vein, Martin Seligman (1970) later claimed that certain species are biologically prepared to learn some behaviors and contraprepared to learn others.

Contiguity and Frequency

These critiques were damaging, but essentially maintained that behaviorist explanations of animal behavior, like behaviorist explanations of human behavior, were more limited in scope than behaviorists had presumed. The Brelands did not think their critique was fatal to behaviorism and suggested that it “should make possible a worthwhile revision in behavior theory” (1961, p. 684). Other critiques questioned the basic explanatory principles of behaviorist learning theories. Theories of classical and operant conditioning, like earlier associationist psychology from Hume to Bain, had assumed that animal and human learning is based upon the principles of contiguity and frequency. As Sigmund Koch put it,

In effect they have given to us as the primary analytic concepts for the most ambitious science ever conceived a mildly camouflaged paradigm for Hume’s analysis of causality.

—(1964, p. 34)

These theories assumed that the strength of a connection between a conditioned stimulus and response (in classical conditioning) or response and reinforcement (in operant conditioning) was a function of the frequency with which a conditioned stimulus is paired (contiguously) with a unconditioned stimulus (in classical conditioning), or a response paired (contiguously) with reinforcement (in operant conditioning), with the optimal temporal interval for pairing being of the order of a fraction of a second.

John Garcia (1917–) challenged these assumptions in a now famous series of studies of conditioned taste aversion (Garcia & Koelling, 1966). Garcia worked for a spell at the Naval Radiation Defense Laboratory in San Francisco, after he failed the statistics course in the psychology department at the University of California at Berkeley (Bolles, 1993). In studying the effects of radiation on rats, he noticed that they would later refuse to drink saccharin if they were exposed to radiation (which made them sick) when they first tasted it. When he returned to Berkeley, Garcia made this the topic of his PhD dissertation. He discovered that rats that refuse to drink saccharin that has been paired with radiation sickness do so after a single trial, violating the principle of frequency, and after an interval of up to 12 hours between drinking saccharin and sickness, violating the principle of contiguity.

Garcia's work demonstrated forms of instinctual preparation and contrapreparation in animal learning, and he co-authored a paper on biological constraints on learning that was included in the 1972 Seligman and Hager collection *The Biological Boundaries of Learning* (Garcia, McGowan, & Green, 1972). However, it took some time for Garcia's work to be generally accepted. When he described the results of his experiments to a learning theorist at the University of California at Los Angeles, he was assured they were impossible (Bolles, 1993), and the mainstream journals rejected his original experimental report (Lubek & Apfelbaum, 1987). However, Garcia's work eventually precipitated a critical reappraisal of theories of conditioning, and he became a distinguished professor of psychology at UC Berkeley.

Garcia's studies indicated that contiguity and frequency are not necessary for conditioned learning. Other studies indicated that contiguity and frequency are not sufficient. Kamin's (1969) studies of "blocking" suggested that prior conditioning to one element of a compound stimulus (such as light in a light and noise complex) attenuates or blocks conditioning to the other element. These and related studies (such as Revusky, 1971) led Rescorla and Wagner (1972) to develop a cognitive explanation of conditioning in terms of discrepancies between anticipated and actual reinforcement, which cast doubt upon the presumed automatic nature of conditioning. As N. J. Mackintosh (1978, p. 54) put it,

Simple associative learning is simple in name only. Animals do not automatically associate all events that happen to occur together. If they did, they would be at the mercy of every chance conjunction of events. In fact, they behave in an altogether more rational manner. By conditioning selectively to good predictors of reinforcement at the expense of poor predictors, and by taking their past experience into account, they succeed in attributing reinforcers to their most probable causes. It is time that psychologists abandoned their outmoded view of conditioning and recognized it as a complex and useful process whereby organisms build an accurate representation of their world.

Consciousness and Conditioning

If this were not bad enough, other researchers suggested that in order for some forms of conditioning to be effective, the link between response and reinforcement must be *consciously* represented. Studies of verbal conditioning had indicated that one could manipulate subjects' use of linguistic items (such as plural nouns) via social reinforcement without their awareness, a form of conditioning commonly known as the **Greenspoon effect** (Greenspoon, 1955). However, Dulany (1968) suggested that, in many of these studies, not only were subjects aware of the relevant response reinforcement connections, but also that such awareness was a necessary condition of successful conditioning. This also cast doubt upon the presumed automaticity of conditioning and undermined the confident denials by Thorndike, Watson, Hull, and Skinner that consciousness plays a role in learning.

Ironically, Skinner's own daughter had pointed this out to him at an early age. In his 1987 book, *Upon Further Reflection*, Skinner reminisced about how he had tried to condition his daughter's foot movements (when she was 3 years old) by rubbing her back:

I waited until she lifted her foot slightly and then rubbed briefly. Almost immediately she lifted her foot again, and again I rubbed. Then she laughed. "What are you laughing at?" I said. "Every time I raise my foot you rub my back!"

—(Skinner, 1987, p. 179)

The Neurophysiology of Learning

Finally, Karl Lashley (1890–1958), Watson's former colleague and one-time supporter of behaviorism (Lashley, 1923), raised critical doubts about the neurophysiological assumptions of traditional behaviorist **peripheralist theories of learning**, which maintained that connections between stimuli and responses are determined independently of "centrally initiated" cognitive cortical processes. Lashley, who held professorships at the universities of Minnesota, Chicago, and Harvard and was later director of the Yerkes Primate Laboratories in Florida, became famous for his work on the neuropsychology of learning in the 1920s. He maintained that there are no cognitive centers of the cerebral cortex governing intelligence and learning comparable to the established sensory and motor centers and that cognitive functions are distributed over the cortex (Lashley, 1929).

At the 1948 Hixon Symposium on "Cerebral Mechanisms in Behavior" held at the California Institute of Technology, he argued that standard behaviorist

accounts of conditioned learning could not accommodate many forms of complex human and animal behavior:

My principle thesis is that . . . input is never into a quiescent or static system, but always into a system which is always already excited and organized. In the intact organism, behavior is the result of interaction of this background of excitation with input from any designated stimulus. Only when we can state the general characteristics of this background of excitation, can we understand the effects of a given input.

—(1951, p. 112)

Lashley claimed that traditional behaviorist explanations in terms of reflex arcs and chains of association or connection could not account for complex serially ordered behavior such as language. He focused on language because it “presents in a most striking form the integrative functions that are characteristic of the cerebral cortex and that reach their highest development in the human thought processes” (Lashley, 1951, p. 113). However, he insisted that these integrative functions, which cannot be explained “in terms of successions of external stimuli,” are also to be found in many other forms of animal and human behavior:

This is true not only of language, but of all skilled movements or successions of movement. In the gaits of a horse, trotting, pacing, and single footing involve essentially the same pattern of muscular contraction in the individual legs. The gait is imposed by some mechanism in addition to the direct relations of reciprocal innervation among the sensory-motor centers of the legs. The order in which the fingers of the musician fall on the keys or fingerboard is determined by the signature of the composition; this gives a *set* which is not inherent in the association of the individual movements.

. . . Not only speech, but all skilled acts seem to involve the same problems of serial ordering, even down to the temporal coordination of muscular contractions in such a movement as reaching and grasping.

—(1951, pp. 116, 121–122)

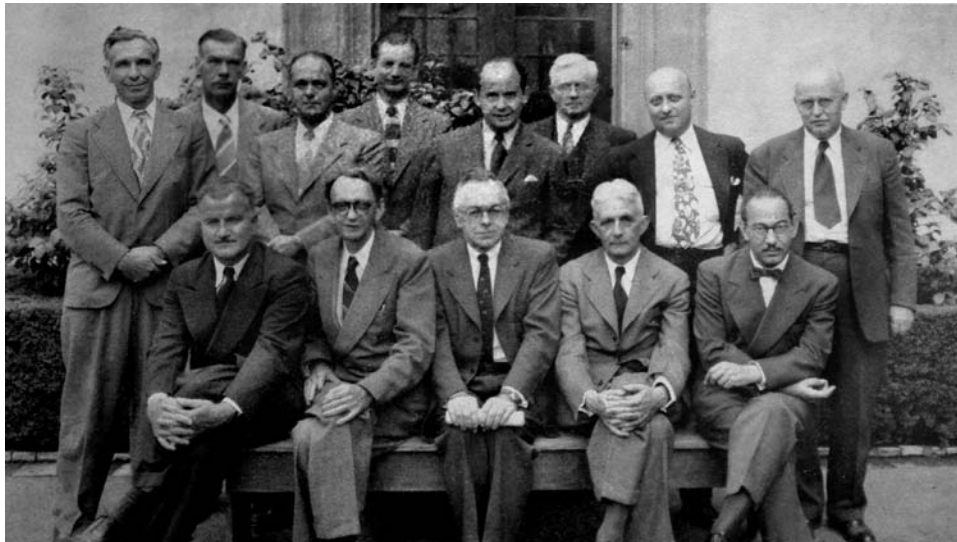
He maintained that many forms of human and animal behavior require explanation in terms of hierarchically organized cognitive control structures that determine complex sequences of serially ordered behavior, which are occasioned by environmental inputs but cannot be explained in terms of associations between environmental inputs and responses or reinforcement histories.

Lashley acknowledged that the forms of explanation he was advocating in terms of central cognitive control processes were analogous to the “determining tendencies” and “ordered thought-processes” postulated by early-20th-century

Würzburg theorists such as Külpe and Selz. In maintaining that cognitive processes in the cerebral cortex play a critical role in the determination of human and animal behavior, Lashley repudiated Watson's famous denial of the causal role of "centrally initiated processes" in the determination of behavior (and vindicated Müller and Bain's 19th-century commitment to "spontaneous" forms of behavior that are products of stored nervous energy in the cerebral cortex).

THE EVE OF THE COGNITIVE REVOLUTION

At the same 1948 Hixon Symposium where Lashley championed cognitive control structures, John von Neumann (1903–1957) and Warren McCulloch (1898–1969) drew parallels between the cognitive processing of information by the cerebral cortex and the newly developed electronic computer (McCulloch, 1951; von Neumann, 1951). Two years later, at the Dartmouth Conference on Learning, neobehaviorist theories of learning were condemned as inconsistent and beset by empirical anomalies (Estes et al., 1954). Six years later, at the 1956 Symposium on Information Theory at MIT, the cognitive revolution in psychology was launched, at least according to Jerome Bruner and George Miller, two of its undisputed leaders (Bruner, 1980; G. Miller, 1989).



Participants at the Hixon Symposium: Karl Lashley (front, second from left), John von Neumann (back, fifth from left), and Wolfgang Köhler (front, fourth from left).

DISCUSSION QUESTIONS

1. Do you agree with the logical positivists that the only things that can be meaningfully stated are those that can be empirically verified? Can you think of claims (including psychological claims) that might be true even if they cannot be empirically tested?
2. If a behavior is purposive—that is, intentionally directed toward a goal or end state—must it be explained in terms of cognition or consciousness? Is it legitimate to explain purposive behavior in terms of instincts or conditioned learning?
3. Tolman held that not all human and animal behavior could be explained in terms of conditioned learning. Did he accept or reject the principle of strong continuity with respect to human and animal psychology?
4. Was Skinner correct in claiming that behavior explained by appeal to anxiety or hunger is sufficiently explained by the external causes of anxiety or hunger? If so, does this mean that explanations of behavior in terms of anxiety or hunger have no independent substantive content?
5. Do you think Chomsky was correct in maintaining that theories of conditioning are incapable in principle of explaining children's acquisition of grammar, given the speed with which they learn grammar on the basis of impoverished data?

GLOSSARY

causally intervening variable A term referencing an internal state of an organism that causally mediates between observable stimuli and behavioral responses. The realist conception of an intervening variable.

cognitivism The view that cognitive theoretical constructs are legitimate and useful in psychological science.

correspondence rule See *operational definition*.

descriptive behaviorism Term employed by B. F. Skinner to describe his form of behaviorism, because he rejected explanatory appeals to unobservable states and processes.

explanatory fiction Term employed by B. F. Skinner to characterize theories about internal cognitive states and processes, which he claimed are vacuous as explanations of relations between observable stimuli and responses and play no role in the development of novel predictions about behavior.

Greenspoon effect The conditioning of verbal behavior through social reinforcement.

human factors research Type of research that focuses on the interaction between human operators and machines, such as torpedoes and anti-aircraft guns, and radar and communication systems.

hypothetical construct Theoretical construct whose meaning is not reducible to empirical laws.

instinctual drift Term employed by Keller and Marian Breland to describe the displacement of learned behavior by instinctual behavior.

intervening variable Theoretical postulate defined in terms of observable independent variables (such as environmental or physiological stimuli) and observable dependent variables (such as behavioral responses).

latent learning Term describing learning in the absence of reinforcement.

logical positivism Form of positivism developed by the Vienna Circle in the 1920s and 1930s, based upon the verification principle.

logically intervening variable A logical device for integrating descriptions of observable stimuli and behavioral responses. The instrumentalist conception of an intervening variable.

mediational theory Type of theory of complex internal response-stimulus (r-s) sequences introduced by neobehaviorists in order to accommodate linguistic behavior and symbolic meaning.

operational definition In logical positivism, the definition of the meaning of a theoretical proposition in terms of observables.

operational behaviorism Term employed by Edward C. Tolman to characterize his form of behaviorism, because of his avowed commitment to the operational definition of theoretical constructs.

operant behavior Emitted behavior whose probability of recurrence is increased by reinforcement.

operant conditioning Form of (instrumental) conditioning that was the focus of B. F. Skinner's research, based upon operant as opposed to respondent behavior.

operational measure Empirical measure of a concept.

operationism Position held by the physicist Percy Bridgman, who maintained that scientific concepts are useful only if there are operational measures of their values.

peripheralist theory of learning Theory of learning in which connections between stimuli and responses are held to be determined independently of "centrally initiated" cognitive cortical processes.

physicalism Version of logical positivism in which observational propositions were held to describe publicly observable properties of physical objects, such as readings on spectrometers or the motion of bodies.

- place versus response controversy** Famous debate between the followers of Tolman and Hull about whether rats running mazes learn cognitive maps or stimulus-response connections.
- psychic machine** Concept of a mechanized robot employed by Hull as a prophylactic against the cognitive interpretation of animal behavior.
- pure stimulus act** According to Hull, an internal response-stimulus (r-s) sequence that causally mediates between an environmental stimulus and a behavioral response.
- purposive behaviorism** Form of behaviorism developed by Edward C. Tolman that focused on purposive or goal-directed behavior.
- radical behaviorism** Form of behaviorism developed by B. F. Skinner based upon operant conditioning, which marked a return to the positivist and inductivist form of behaviorism developed by John B. Watson.
- respondent behavior** Behavior elicited by unconditioned or conditioned stimuli.
- schedules of reinforcement** The variety of fixed and variable interval and ratio schedules of reinforcement employed by B. F. Skinner in his study of conditioned learning.
- scientific empiricism (logical empiricism)** Later form of logical positivism based upon physicalism that neobehaviorists embraced.
- sensationalism** Version of logical positivism in which observational propositions were held to describe the properties of private sense experience, such as the intensity of colors or apparent differences in weight.
- shaping** Method of operant conditioning developed by B. F. Skinner in which a target behavior is produced through the reinforcement of progressive approximations to that behavior.
- surplus meaning** The meaning of theoretical postulates that is additional to or independent of operational definition in terms of empirical laws.
- verification principle** The logical positivist principle that the only meaningful factual propositions are those verifiable by observation.
- theoretician's dilemma** Conceptual dilemma created by insistence on the exhaustive operational definition of theoretical postulates, which implies their dispensability.

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The Cognitive Revolution

THE “COGNITIVE REVOLUTION” IN PSYCHOLOGY EMERGED FROM postwar developments in information theory and computer science. The development of the electronic computer created a new and technically proven model of the mind as a mechanical information processor, conceived of as operating on the same sorts of “rules and representations” employed by “intelligent” machines (Bechtel, 1988).

One of the peculiarities of the cognitive revolution was that many of its pioneers came to conceive of their own intellectual achievement in terms of Thomas Kuhn’s (1970) analysis of the structure of scientific revolutions, according to which one general theoretical or methodological paradigm is replaced by a radically different paradigm, under the pressure of accumulating empirical anomalies (Lachman, Lachman, & Butterfield, 1979). Kuhn’s influential *The Structure of Scientific Revolutions* was first published in 1962, as the new forms of cognitive psychology were being advanced and developed by Jerome Bruner (1915–), George Miller (1920–), Ulric Neisser (1928–), Allen Newell (1927–1992), and Herbert Simon (1915–2001). As James J. Jenkins (1923–) later remarked, during the early years of the cognitive revolution in psychology, “everyone toted around their little copy of Kuhn’s *The Structure of Scientific Revolutions*” (Jenkins, quoted in Baars, 1986, p. 249).

Although there was no revolution in the strict Kuhnian sense, the development of cognitive theories from the 1950s and 1960s onward did mark a genuine discontinuity with behaviorist theories, including later “liberalized” neobehaviorist theories in terms of internal “mediating” r-s sequences (N. Miller, 1959; Osgood, 1957). Although the primary stimulus for the cognitive revolution came from without, the empirical problems faced by neobehaviorism in the 1950s and 1960s created an intellectual climate that left many psychologists predisposed to theoretical and methodical change. As Jenkins put it, “things were boiling over . . . a new day was coming” (Jenkins, quoted in Baars, 1986, p. 249).